Adding fermionic dimension-8 operators to SmeftFR v4

As dimension-8 fermionic operators are very numerous, we did not include full set of them as a standard feature of published code, but we provide instructions on how to do that, illustrating it below on a chosen example of fermionic operator. We are working in the basis of [1]. Below we list the steps which one should follow.

- Define the Mathematica code for the operator itself. We present this on the example from Table 7 in Ref. [1]:

$$Q^{(4)}_{e^2WH^2D} = (\bar{e}_p \gamma^\nu e_r) \left(H^\dagger \overleftrightarrow{D}^{I\mu} H \right) \widetilde{W}^I_{\mu\nu}$$

  The corresponding definition should be placed in file `Lagrangian/215_psiXphi2D.fr` (we follow the numeration of operator classes in [1] - operator from class XX should be defined in the file `lagrangian/2XX_class_name.fr`):

```
LQe2Wphi2Dn4 := Module[{sp1,sp2,m,ii,jj,ff1,ff2,mu,nu,al,be,aux},
aux = 2 Ta[m,ii,jj] (Phi8bar[ii] DC[Phi8[jj],mu] - DC[Phi8bar[ii],mu] Phi8[jj])*
lRbar[sp1,ff1].lR[sp2,ff2] Ga[nu,sp1,sp2] Eps[mu,nu,al,be]/2 HC[FS[Wi,al,be,m]];
aux = ExpandIndices[ ToExpression[SMEFT$WB <> "e2Wphi2Dn4"][ff1,ff2] aux,
                    FlavorExpand->{SU2W,SU2D} ];
aux /.  SMEFTGaugeRules ];
```

  We follow standard FeynRules syntax (see SM Lagrangian definition included as an example in FeynRules distribution), with the exception of replacement of `Phi` $\to$ `Phi8` used to speed up expansions in mass dimensions. For consistency and to avoid accidental name repetitions, naming of operators in code should follow their naming in [1] (preceded by "L" as "Lagrangian"), in this case $Q^{(4)}_{e^2WH^2D} \to$ `"e2Wphi2Dn4"` (in the operator names we also replace `H` $\to$ `phi`), with Lagrangian entry named `LQe2Wphi2Dn4`, extra `"LQ"` always preceding operator name.

- Open file `/code/smeft_variables.m` and:

  1. Add operator name `"e2Wphi2Dn4"` to the relevant list near the top of file, their names are self-explanatory:
     `SMEFT$Dim8TwoFermionOperators`,
     `SMEFT$Dim8FourLeptonOperators`,
     `SMEFT$Dim8FourQuarkOperators`,
     `SMEFT$Dim8TwoQuarkTwoLeptonOperators`,
     `SMEFT$Dim8FourFermionBLVOperators`.

     In our example obviously one should include it in
     `SMEFT$Dim8TwoFermionOperators={...,"e2Wphi2Dn4",...}` .
  2. Add new line to the `SMEFT$TensorWC` table:

```
SMEFT$TensorWC = {
            ...
            {"e2Wphi2Dn4",{VLR,VLR},False,2},

            ...}
```

The syntax of the extra lines in `SMEFT$TensorWC` consist of:

1^{st} entry: operator name

2^{nd} entry: list of fermionic rotation matrices. They are denoted as `VXY`, where `X=(L,U,D)` is the fermion type and `Y` its chirality. For instance, for operator containing fermionic current $l_L u_R$ one should add as a 2^{nd} entry {`VLL, VUR`}. Similar entry for 4-fermion $q_L q_L d_R d_R$ operator would be {`VDL, VDL, VDR, VDR`}, and so on. For left $SU(2)$ doublet q_L either `VUL` or `VDL` can be used, it affects in which places CKM matrix will appear in fermion mass basis interactions.

3^{rd} entry: is the operator B or L violating? Always `False` for 2-fermion dimension-8 operators.

4^{th} entry: operator "symmetry class". Here for details see comments on index symmetries of WCs of dimension-6 fermionic operators ("Straub classes") in file `code/smeft_variables.m`, above the definition of `TensorInd` table. For dimension-8 fermionic operators, one can usually assign for them one of the classes defined for dimension-6 case. If flavor index symmetries for newly added operator are not known or differ from dimension-6 ones, one can assign class 1 for 2-fermion and class 9 for 4-fermion case, corresponding to no symmetries imposed (they can still be applied later at numerical level, initializing values of WCs). Experienced and brave user can also define new types of symmetry classes, modifying appropriately `TensorInd` table and `TensorWCSymmetrize` routine in `code/smeft_functions.m` file.

- Go to `smeft_fr_init.m` (or `SmeftFR-init.nb`) and add operator name (here `"e2Wphi2Dn4"`) to the `OpList8` which lists dimension-8 operators included in calculations.

References

[1] C. W. Murphy, Dimension-8 operators in the Standard Model Effective Field Theory, JHEP 10 (2020) 174. `arXiv:2005.00059`, `doi:10.1007/JHEP10(2020)174`.